Proof of LOOCV $CV^n_{Err} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1-h_{ii}}\right)^2$

Consider linear regression $Y = X\beta + \varepsilon$ (Y: $n\times 1$, X: $n\times p$, β: $p\times 1$, ε: $n\times 1$)

$\hat{\beta} = (X^TX)^{-1}X^Ty$ $\qquad$ X_i^T is $1\times p$

$H = X(X^TX)^{-1}X^T$

Let X_i^T be the ith row of X

$X^{(-i)}$ and $Y^{(-i)}$ be design matrix and outcome with ith row removed

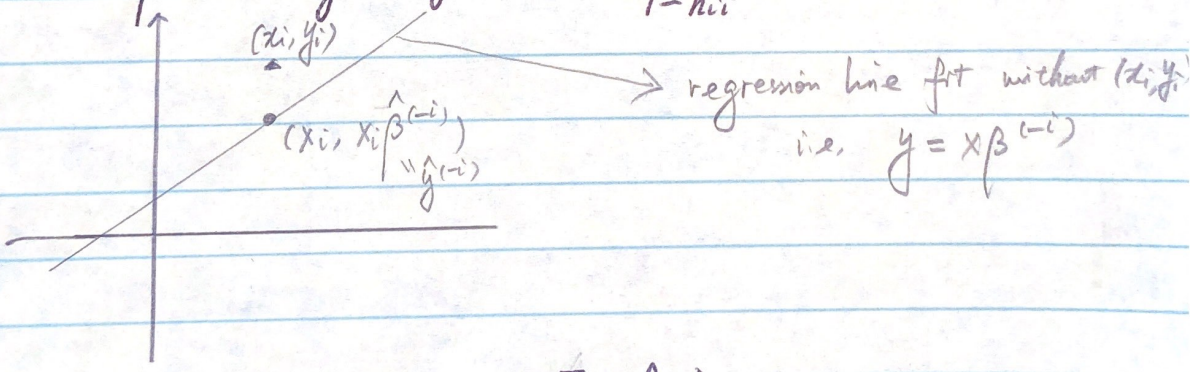
Note $Hy = X\hat{\beta} = \hat{y}$ $\quad$ ($H\hat{y} = X(X^TX)^{-1}X^Ty$, with $\hat{\beta}$)

$$\Rightarrow \sum_{j=1}^{n} h_{rj}\, y_j = \hat{y}_r \qquad (1)$$

Let $H^{(-i)} = X^{(-i)}\left[X^{(-i)T}X^{(-i)}\right]^{-1}X^{(-i)T}$ $\quad$ ith row removed from H

$$\hat{\beta}^{(-i)} = \left[X^{(-i)T}X^{(-i)}\right]^{-1}X^{(-i)}Y^{(-i)}$$

want to prove $y_i - \hat{y}^{(-i)} = \dfrac{y_i - \hat{y}_i}{1-h_{ii}}$



augment $X^{(-i)}$, $Y^{(-i)}$ with X_i^T, $\hat{y}^{(-i)}$

i.e., $X(a) = X^{(-i)}$ with X_i^T added back $= X$

$$Y(a) = \begin{pmatrix} y_1 \\ \vdots \\ y_{i-1} \\ \hat{y}^{(-i)} \\ y_{i+1} \\ \vdots \\ y_n \end{pmatrix} \Rightarrow \hat{\beta}(a) = \hat{\beta}^{(-i)}$$

$H(a) = H$

i.e., since $(X_i, \hat{y}^{(-i)})$ is on the regression line augmenting $X^{(-i)}$ and $Y^{(-i)}$ with X_i^T and $\hat{y}^{(-i)}$ will not change LOOCV estimate $\hat{\beta}^{(-i)}$

rewrite ① $\Rightarrow$ $\hat{y}^{(-i)} = \sum_{j \neq i} h_{ij} y_j + h_{ii} \hat{y}^{(-i)}$

$H(a) Y(a) = \hat{Y}(a)$

$$= \sum_{j=1}^{n} h_{ij} y_j - h_{ii} y_i + h_{ii} \hat{y}^{(-i)}$$

// using ① again

$$= \hat{y}_i - h_{ii} y_i + h_{ii} \hat{y}^{(-i)}$$

$$\Rightarrow y_i - \hat{y}^{(-i)} = y_i - (\hat{y}_i - h_{ii} y_i + h_{ii} \hat{y}^{(-i)})$$

$$\Rightarrow y_i - \hat{y}^{(-i)} = (y_i - \hat{y}_i) + h_{ii} (y_i - \hat{y}^{(-i)})$$

$$\Rightarrow y_i - \hat{y}^{(-i)} = \frac{y_i - \hat{y}_i}{1 - h_{ii}}$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}^{(-i)})^2 = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2$$